Case 6 — Mr. George Miller

75 years old · Male Parkinson's Disease · 12 years duration Hoehn & Yahr Stage IV (Advanced) Motor fluctuations · Two falls (turning) · Shuffling gait · FOG

Neurological Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Freezing of gait (FOG) — turning-triggered, confirmed by falls context
- Shuffling gait — reduced step length, increased cadence, poor heel clearance
- On-off motor fluctuations — sudden loss of mobility unpredictably
- No ROM deficits — impairment is central/neurological, not musculoskeletal

■■ RED FLAGS

- 1. Recurrent injurious falls — already present
- 2. Severe orthostatic hypotension or syncope
- 3. Signs of aspiration or significant dysphagia
- Significance: Bulbar dysfunction in advanced PD impairs swallowing coordination; aspiration pneumonia is the leading cause of death in advanced PD. Reduced cough reflex may mean aspiration is silent.
- 4. Acute cognitive deterioration, confusion, or delirium
- Action: Suspend assessment; notify medical team; basic cognitive screen (abbreviated MMSE); check for signs of infection; do not attempt any exercise or mobility work until medically cleared.
- 5. Signs of respiratory compromise (reduced chest expansion, low SpO2, tachypnoea)
- Significance: Thoracic rigidity, kyphosis, and potential aspiration reduce respiratory reserve in advanced PD; acute decompensation is life-threatening.

■ PATHOPHYSIOLOGY (KEY POINTS)

- This explains why FOG is most dangerous in confined, complex environments.
- Motor fluctuations — the "on-off" phenomenon — arise because after 12 years of levodopa therapy, the remaining dopaminergic neurons have lost their capacity to buffer and gradually release dopamine.
- Dopamine availability in the striatum now mirrors levodopa plasma levels directly, oscillating sharply with each dose cycle.
- When levodopa levels fall (wearing-off) or fluctuate unpredictably (on-off), the striatum loses dopaminergic tone abruptly, and the patient transitions from functional mobility to near-immobility within minutes.
- This sudden transition — combined with FOG and impaired postural reflexes — is what makes advanced PD so dangerous from a falls perspective.
- Miller's 12-year disease duration and Hoehn & Yahr IV status are consistent with the degree of dopaminergic depletion and Lewy body spread expected at this stage.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Key Neurological Observations (Before Formal Testing)

Gait observation: Observe during entry to clinic — note shuffling pattern, heel clearance, step length, cadence; does FOG occur at the doorway? How does he manage the turn to sit down? Is there festination (involuntary acceleration)? Postural stability: Standing posture — degree of thoracic kyphosis and forward head position; sway; does he need to hold furniture? Centre of mass position relative to base of support Tone:

Freezing of Gait Questionnaire (FOG-Q) + Timed 360° Turn Test

What it measures: How to perform (360° Turn): Ensure a safety spotter is present — stand close enough to catch without interfering Patient stands on a marked spot, feet shoulder-width apart Instruct: "Turn a full circle to the right, then a full circle to the left. Walk at your usual pace."

Pull Test (Postural Instability — MDS-UPDRS Item 3.12)

What it measures: Automatic postural righting responses — whether Mr. Miller can recover his balance from an unexpected backward perturbation. At Hoehn & Yahr IV, I expect severely impaired postural reflexes. This test directly determines the severity of postural instability, guides walking aid prescription, informs carer training in safe assistance, and is required for accurate UPDRS staging. How to perform: Perform with a second person present as a safety catcher at the front — do NOT perform alone in advanced PD Patient stands with feet shoulder-width apart; warn them: "I'm going to pull your shoulders backwards firmly — try to stay balanced, you may take steps"

■ TREATMENT PLAN

- Does Mr. Miller understand the relationship between medication timing and safe mobility? Lack of understanding leads to dangerous unsupervised activity during "off" periods
- Test 3: Dual-Task Walking Assessment (10-Metre Walk — Single vs. Dual Task)
- What it measures:
- Measure a clear 10-metre walkway; mark start and stop lines
- Single task: ask patient to walk at comfortable pace end-to-end; record time and step count

• Dual task: repeat the walk while patient simultaneously counts backwards from 50 in 3s, or names alternating fruits and vegetables
• Calculate DTC = [(single-task speed – dual-task speed) / single-task speed] × 100%
• Also note whether FOG is triggered more easily during the dual-task condition
• Positive / Abnormal finding:
• ■ Outcome Measures
• MDS-UPDRS Part III (Motor Examination):
• Berg Balance Scale (BBS):